

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Reinforcement Learning [NPT_4126]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Describe Reinforcement Learning (RL) in your own words and compare it with supervised learning by highlighting how the learning process differs between the two. (2)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Discuss the difference between supervised, unsupervised, and reinforcement learning with suitable examples. (3)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- C) Apply the Temporal Difference (TD(0)) learning concept to a simple Markov chain with three weather states, Sunny, Cloudy, and Rainy. Define the possible rewards and transition probabilities, and explain how TD(0) would update the estimated value of each state after observing one episode (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Describe the key components of a Markov Decision Process ? (2)
- A) Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Distinguish between value-based and policy-based reinforcement learning approaches with suitable examples. (3)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Consider an example of a Markov Decision Process representing a self-driving car at intersections. Interpret its states, actions, transition probabilities, and rewards. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Given a small Markov Decision Process, interpret how policy evaluation can be used to update the value of a state after one iteration. (2)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Given a grid-world example, demonstrate one iteration of policy iteration, showing policy improvement and evaluation steps. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Compare the gradient-based optimization in policy gradient methods with value-based approaches like Q-learning. What advantages does the former offer? (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4) Examine how the λ parameter in TD(λ) influences the trade-off between bias and variance in learning. (3)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Compare linear and non-linear function approximation methods in reinforcement learning, highlighting their strengths and limitations. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Consider a reinforcement learning task with complex, high dimensional states. Examine whether linear function approximation or neural network-based approximation is more suitable and justify your choice. (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Demonstrate how an agent can use function approximation (e.g., linear or neural network) to estimate the value function. (3)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Compare POMDPs with standard MDPs, focusing on how uncertainty in observations impacts decision-making. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Apply the concept of policy gradients combined with deep learning to explain how reinforcement learning can be used in a real-world application such as healthcare or data science. Provide a suitable example to illustrate the implementation. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4126:Reinforcement Learning

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	15
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4126:Reinforcement Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	7(14%)
3	Application	6	22(44%)

4	Analysis	6	21(42%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)